

Giang Do (James)

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Google Scholar

RESEARCH INTERESTS

My research focuses on the efficiency, scalability, and reliability of large foundation models. In particular, I work on **Sparse Mixture-of-Experts (MoE)** for training and inference of large language models—studying routing, representational collapse, and the emergence of specialised experts—and on **controllable and robust reasoning** in LLMs. A unifying thread of my PhD is the *binding problem*: how foundation models compose, separate, and route information across modular components.

EDUCATION

- Deakin University** VIC, Australia
• *PhD in Information Technology* Jun 2024 – Present
Thesis topic: Improving Foundation Models by Addressing the Binding Problem.
Supervisors: Prof. Truyen Tran, Dr. Hung Le (Applied AI Institute, A²I²).
- The University of Tennessee at Chattanooga** TN, USA
• *M.Sc. in Computer Science; GPA: 4.0/4.0 (Class rank: 1st)* Jan 2023 – May 2024
Selected coursework: Mathematical Statistics, Machine Learning, Advanced Systems Software.
- VNU University of Science** Hanoi, Vietnam
• *M.Sc. in Data Science; GPA: 3.75/4.0 (Class rank: 2nd)* Dec 2020 – Dec 2022
Thesis: Identifying Constructive Comments for Vietnamese News (8.2/9.0; best thesis award).
Supervisor: Dr. Tuan-Anh Hoang, RMIT Vietnam University.
- Hanoi University of Science & Technology (HUST)** Hanoi, Vietnam
• *B.Tech. in Information Technology; GPA: 3.36/4.0 (Top 7%)* Aug 2019 – Jun 2021
Thesis: Constructive Comment Classification (9.25/10.0; best thesis award).
- Foreign Trade University** Hanoi, Vietnam
• *B.A. in International Business Economics; Classification: Good* Aug 2010 – Jun 2014

PUBLICATIONS

(**Bold** = author. * denotes first author.)

Refereed Conference & Journal Papers

- Eigenvectors of Experts are Training-free Non-collapsing Routers**
Giang Do, Hung Le, Truyen Tran*
ICML 2026 (Spotlight).
- Rethinking Sparse Mixture of Experts from a Unified Perspective**
Giang Do, Hung Le, Truyen Tran*
ICML 2026.
- Do Domain-specific Experts Exist in MoE-based LLMs?**
Giang Do, Hung Le, Truyen Tran*
ACL 2026 (Findings).
- SimSMoE: Toward Efficient Training of Mixture of Experts via Solving Representational Collapse**
Giang Do, Hung Le, Truyen Tran*
NAACL 2025 (Findings).
- On the Role of Discrete Representation in Sparse Mixture of Experts**
Giang Do, Kha Pham, Hung Le, Truyen Tran*
Transactions on Machine Learning Research (TMLR), 2025.
- HyperRouter: Towards Efficient Training and Inference of Sparse Mixture of Experts**
Giang Do, Khiem Le, TrungTin Nguyen, Quang Pham, Binh T. Nguyen, Thanh-Nam Doan, Chenghao Liu, Savitha Ramasamy, Xiaoli Li, Steven Hoi*
EMNLP 2023 (Main Conference).

Preprints & Under Review

- Measuring and Improving Robustness of Large Reasoning Models under Task Interruption**
Giang Do, Hung Le, Truyen Tran*
Under review, 2026.
- Think in Episodes: Controlling Reasoning in Language Models via Episodic Inference**
Giang Do, Hung Le, Truyen Tran*
Under review, 2026.

9. CompeteSMoE: Effective Training of Sparse Mixture of Experts via Competition

Quang Pham, **Giang Do**, Huy Nguyen, TrungTin Nguyen, Chenghao Liu, Mina Sartipi, Binh T. Nguyen, Savitha Ramasamy, Xiaoli Li, Steven Hoi, Nhat Ho

Under review. [arXiv:2402.02526](#).

RESEARCH EXPERIENCE

- **Applied Artificial Intelligence Institute (A²I²), Deakin University** VIC, Australia
Research Assistant Oct 2024 – Present
 - **Radiology Report Generation Evaluation:** Surveyed existing evaluation methods for radiology report generation and proposed a knowledge-graph-based metric better aligned with expert clinical judgement.
 - **3D Infant Modelling and Video Tracking:** Developed 2D detection and 3D tracking pipelines for infant videos to support downstream behavioural and clinical analysis.
- **Center for Urban Informatics and Progress (CUIP), UT Chattanooga** TN, USA
Graduate Research Assistant Jan 2023 – May 2024
 - **Large Language Models:** Investigated hypernetworks and Sparse Mixture-of-Experts to improve training stability and efficiency of large language models (Transformer-XL, RoBERTa). This work led to the EMNLP 2023 *HyperRouter* paper.
 - **Camera-LiDAR Data Fusion:** Studied state-of-the-art 2D and 3D object detection models for fusing camera and LiDAR data in urban-scene perception.

INDUSTRY EXPERIENCE

- **Panasonic R&D Center Vietnam** Hanoi, Vietnam
AI Project Leader Feb 2021 – Jan 2023
 - **Digital Twin for Buildings:** Led a team of 4 engineers to design and deliver a digital-twin system for building modelling and monitoring (3D BIM + Unity).
 - **3D Object Detection:** Trained and fine-tuned state-of-the-art 3D object detectors, achieving +5% accuracy over pre-trained baselines.
 - **Unsupervised Ceiling-Light Recognition:** Proposed an unsupervised point-cloud method that recognised ceiling lights with >90% accuracy.
 - **Semantic 3D Mapping:** Designed an algorithm mapping 2D instance segmentation outputs onto 3D building models.
- **Techcombank (Vietnam Technological & Commercial Joint Stock Bank)** Hanoi, Vietnam
Senior Data Scientist Oct 2019 – Jan 2021
 - **Pricing Models:** Built ML systems to recommend bond/stock buy-sell prices for retail clients.
 - **Anomaly Detection:** Developed models to detect abnormal logins and trading patterns for security and compliance.
- **Toyota Motor Vietnam** Hanoi, Vietnam
Data Analyst Jan 2016 – Jul 2019
 - **Sales Forecasting:** Built time-series forecasting models informing production and marketing decisions for top management.

HONORS AND AWARDS

- Spotlight Paper, ICML 2026 (*Eigenvectors of Experts are Training-free Non-collapsing Routers*).
- UTC Outstanding Graduate Student Award, 2023–2024.
- Best Master's Thesis Award, VNU University of Science — “Identifying Constructive Comments for Vietnamese News”, Dec 2022.
- Best Employee Award, Panasonic R&D Vietnam, Jul 2022.
- Best Project Award, Panasonic R&D Vietnam, Dec 2021.
- Best Bachelor Thesis Award, HUST — “Constructive Comment Classification”, Jun 2021.
- Vingroup Science and Technology Scholarship for M.Sc. Data Science students, 2021–2022.

ACADEMIC SERVICE

- Reviewer: AISTATS 2026, NeurIPS 2026.

TECHNICAL SKILLS

- **Programming:** Python (primary), R, SQL, C/C++, Bash
- **ML / DL Frameworks:** PyTorch, HuggingFace Transformers, DeepSpeed, TensorFlow, scikit-learn
- **Infrastructure:** Linux, Slurm, Docker, Git; multi-GPU training (A100/H100)

REFERENCES

Prof. Truyen Tran

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Applied Artificial Intelligence Institute (A²I²)
Deakin University
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Dr. Hung Le

Senior Lecturer
Applied Artificial Intelligence Institute (A²I²)
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